

Capacitor Charging and a Power On Reset for an Active Low Microcontroller

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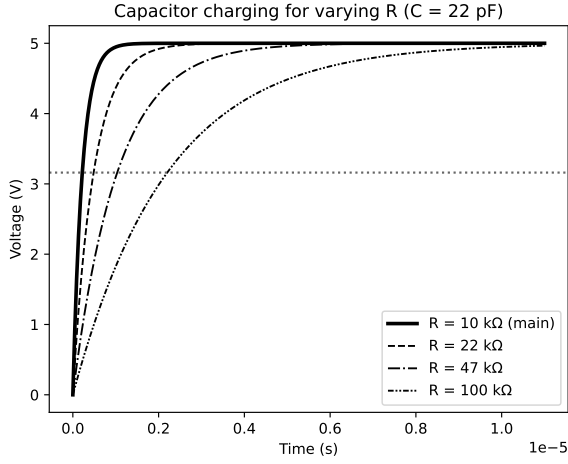


Fig. 1. Charging curves for four resistances, $C = 22\text{pF}$.

Abstract—This document derives the charging behaviour of a resistor-capacitor (RC) network, the voltage reached after one time constant, and applies the result to a power-on-reset (POR) circuit for microcontroller with an active-low reset.

I. INTRODUCTION

When a capacitor C is charged through a series resistor R from a constant supply V_0 , the voltage rises according to

$$V(t) = V_0 \left(1 - e^{-t/RC}\right), \quad (1)$$

where $\tau = RC$ is the time constant. Substituting $t = \tau$ gives $V(\tau) = V_0(1 - e^{-1}) \approx 0.632V_0$

II. POWER ON RESET WITH AN ACTIVE LOW RESET

A power-on-reset circuit exploits(1) directly: R from V_{CC} to the reset node and C from that node to ground hold the microcontroller's active-low reset pin low until the node crosses V_{IH} , releasing the device.

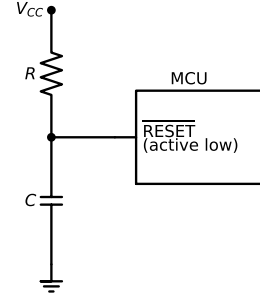


Fig. 2. power-on-reset network driving an active-low reset pin.

A.sizingRforaTimingConstraint

The reset pin releases the instant the node voltage reaches V_{IH} . Setting $V(t_{release}) = V_{IH}$ in (1) and solving for $t_{release}$:

$$\begin{aligned} V_{IH} &= V_0 \left(1 - e^{-t_{release}/RC}\right) \\ \frac{V_{IH}}{V_0} &= 1 - e^{-t_{release}/RC} \\ -t_{release}/RC &= 1 - \frac{V_{IH}}{V_0} \\ -\frac{t_{release}}{RC} &= \ln \left(1 - \frac{V_{IH}}{V_0}\right) \\ t_{release} &= -RC * \ln \left(1 - \frac{V_{IH}}{V_0}\right) \end{aligned} \quad (2)$$

Rearranging (2) for R , given target $t_{release}$ and chosen C :

$$R = \frac{-t_{release}}{C * \ln \left(1 - \frac{V_{IH}}{V_0}\right)} \quad (3)$$

1) *Worked example:* Take $V_{IH} = 0.7V_0$, $V_0 = 5.0V$, and a target $t_{release} = 10\text{ms}$. The $C = 22\text{pF}$ used for the plots above was chosen only to keep the curves readable at a sub-microsecond timescale; a practical timing design instead starts from $C = 100\text{nF}$. Evaluating denominator of (3):

$$\ln(1 - 0.7) = \ln(0.3) \approx -1.204. \quad (4)$$

Substituting into (3):

$$R = \frac{-(10 * 10^{-3})}{(100 * 10^{-9}) * (-1.204)} \approx 83\text{k}\Omega \quad (5)$$

An $83\text{k}\Omega$ resistor with a 100nF capacitor gives a reset hold-off of approximately 10ms .

III. APPLICATIONS AND LIMITATIONS

A. Part A

- A specific device that is an appropriate design choice where the RC only power-on-reset circuit is a standalone digital desktop alarm clock based on an 8-bit microcontroller.
- The microcontroller only needs a brief power-on delay to allow V_{CC} to stabilize before execution initialization routines. High-accuracy startup timing is not needed.
- Resistor and Capacitor Requirements:
 - Resistor (R): $100k\Omega$
 - Capacitor (C): $100nF$
 - Supply Voltage (V_0): $5.0V$
 - Input-High Threshold Voltage (V_{IH}): $0.7V_0 = 3.5V$

$$R = \frac{-t_{release}}{C * \ln\left(1 - \frac{V_{IH}}{V_0}\right)}$$

$$t_{release} = -R * C * \ln\left(1 - \frac{V_{IH}}{V_0}\right) \quad (6)$$

$$t_{release} = 100k * 100n * -1.204$$

$$t_{release} = 0.01204s \approx 12.04ms$$

B. Part B

Although power-on-reset circuit is low costing and simple, it suffers from many limitations:

- 1) Component Tolerance: Standard variations in resistors and capacitor values combine linearly causing up to a 20% variance in reset timings
- 2) Slow Supply Ramp: If V_{CC} rises slowly, the reset voltage will linger near the threshold voltage causing the microcontroller to execute code before the voltage stabilizes.
- 3) No Software or Watchdog Triggering: The network is passive, meaning it cannot be re-triggered by a watchdog timer or a internal software command to force a system reboot if the application hangs.

C. Part C

When small variants ΔR and ΔC occur in components, the new release time becomes:

$$t + \Delta t = k(R + \Delta R)(C + \Delta C) = kRC + kR\Delta C + kC\Delta R + k\Delta R\Delta C$$

$$t + \Delta t \approx kRC + kR\Delta C + kC\Delta R$$

$$\Delta t \approx kR\Delta C + kC\Delta R \quad (7)$$

Dividing by $t = kRC$ yields the relative timing error formula:

$$\frac{\Delta t}{t} \approx \frac{\Delta R}{R} + \frac{\Delta C}{C} \quad (8)$$

Using the values selected in Part A:

- R : $100k\Omega$ with a standard 5% tolerance
- C : $100nF$ with a standard 10% tolerance
- Voltage Ratio: 1.204

- $t_{release} = 12.04ms$

Minimum Case:

$$R_{min} = 100k\Omega * 0.95 = 95k\Omega$$

$$C_{min} = 100nF * 0.90 = 90nF \quad (9)$$

$$t_{min} = (1.204) * (95k) * (90n) = 10.29ms$$

Maximum Case:

$$R_{max} = 100k\Omega * 1.05 = 105k\Omega$$

$$C_{max} = 100nF * 1.10 = 110nF \quad (10)$$

$$t_{max} = (1.204) * (105k) * (110n) = 13.91ms$$

Due to component tolerances, the actual reset release time will fall within the range of 10.29 ms to 13.91 ms. This is about a 15% tolerance around $t_{release} = 12.04$ ms.